

Comparison Between Spam/Ham Detector and Phishing Detector

1. Definitions

Spam/Ham Detector: A system that classifies emails into Spam (unwanted or promotional messages) and Ham (legitimate normal emails). Its primary goal is filtering unwanted messages.

Phishing Detector: A security-focused system that detects fraudulent emails or URLs designed to steal sensitive information such as passwords, banking details, or personal data.

2. Examples

Spam Example: "Buy cheap products now! Limited time discount!"
Ham Example: "Meeting scheduled at 10 AM tomorrow."

Phishing Example: "Your bank account has been locked. Click here to verify your password immediately."
Legitimate Example: Official email from a verified domain without suspicious links.

3. Tabular Comparison

Feature	Spam/Ham Detector	Phishing Detector
Main Goal	Filter unwanted emails	Prevent fraud and data theft
Threat Level	Mostly annoying	Highly malicious and dangerous
Focus Area	Promotions, bulk messages	Credential theft, fake URLs, impersonation
URL Analysis	Rarely used	Very important
Email Header Analysis	Limited	Important
Risk Scoring	Not included	Included in advanced systems
Explainability	Usually not required	Highly important
Category Classification	Not required	Credential scam, impersonation, malware
Security Impact	Low	High (financial and data risk)